

“Different Approaches for Pothole Detection”– A Review

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Abstract— In the era of increasing road accidents intensively, Road anomaly detection is crucial. It not only ensures traffic safety but also optimizes infrastructure maintenance, and improves navigation of autonomous vehicle. In this review paper, recent advancements in machine learning (ML), deep learning (DL) are explored for detecting road anomalies such as potholes, cracks, debris, and other surface irregularities. Genetic Algorithm, Isolation Forest, XGBoost etc. machine learning approaches, along with MaskRCNN, CMNN-RADC, YOLOv5, YOLOv7 etc. deep learning models have been critically and comparatively analyzed in the context of road anomaly detection. Furthermore, ensembled autoencoder, TinyML with edge computing, Segformer network like hybrid approaches have also been sightseen. Out of all the approaches, hybrid approaches are identified with higher accuracy – 80+% to detect road anomalies. In the future, more hybrid approaches can be developed and examined using YOLO inspired custom CNN for getting higher accuracies and more simplified work for detecting road anomalies.

Keywords—Road Anomaly Detection; Machine Learning; Deep Learning; Machine Learning; Pothole Detection.

I. INTRODUCTION

A. Research Background

In the recent years, the vitality of the road safety has been highlighted due to the rapid transformation of transportation systems. Real-time detection of the road anomalies such as accidents, unexpected obstacles, defective road surfaces etc. is essential to prevent accidents, facilitate efficient traffic flow, and enable proactive road infrastructure maintenance [1].

B. Problem Definition

The Global Status Report on Road Safety 2015 states that 50 million people get non-fatal injuries and more than 1.2 million people die on the world's roadways annually. Surprisingly, while accounting for just roughly 54% of all registered automobiles worldwide, low- and middle-income nations account for nearly 90% of all traffic accidents [14].

India, in particular, is experiencing a severe road safety crisis, with 400 fatalities and 1,374 accidents every day on average. This is equivalent to about 17 fatalities and 57 accidents each hour. According to projections, India may see close to 241,751 road accident-related deaths by 2030 if current trends continue and strong road safety measures are not put in place [15].

In order to lessen the financial and human costs associated with road anomalies, these figures highlight the urgent need for cutting-edge road safety technologies, such as real-time anomaly detection and predictive safety systems [16]

C. State-of-the-art

The advanced machine learning, deep learning, and hybrid approaches are the promising solutions for detecting diverse road anomalies. ML approaches such as Genetic Algorithm, Isolation Forest, XGBoost etc., deep learning approaches including MaskRCNN, CMNN-RADC, YOLOv5, YOLOv7 etc., and hybrid approaches ensembled autoencoder, TinyML with edge computing, Segformer network etc. are contributing accuracy range 80 – 95% to detect road anomalies. By utilizing data from several sources, including cameras, LiDAR, and radar, these techniques enhance the system's capacity to detect irregularities even under difficult circumstances. Additionally, there are now more readily available datasets related to anomaly detection and road safety, which serve as a basis for developing and accessing machine learning models.

D. Scope of the paper

The scope of this paper is limited to provide comprehensive and comparative existing approaches of road anomaly detection. It focuses on methodologies, algorithms, datasets, and performance metrics, followed by different researchers, along with the merits and demerits of their approach to detect road anomalies. It also examines future directions to ensure safer and smarter road anomaly detection systems.

II. REVIEW METHODOLOGY

This paper follows a systematic literature review methodology to conduct comprehensive and comparative literature review. It follows a specific search strategy, a criterion to include and exclude research papers, and a categorization framework to conduct literature review [6].

A. Search strategy

To collect reputed, relevant, and reliable research papers, IEEE Xplore, ScienceDirect, SpringerLink, Elsevier, and Google Scholar, like databases have been followed.

From where the research papers published within 2020 and 2025 are collected using the keywords such as

1. “Pothole Detection” using “Machine Learning” OR “Deep Learning” OR “CNN” OR “YOLO”
2. “Road Anomaly Detection” using “Machine Learning” OR “Deep Learning” OR “CNN” OR “YOLO” [3]

B. Inclusion and exclusion criteria

To refine the list of the collected research articles, only relevant papers along with the experimental validation, recent publication years, topic relevance etc. have been

shortlisted. Out of 50, only 25 papers are considered for the literature review [4].

C. Categorization framework

After searching and shortlisting the research papers, they are categorized into “Machine Learning based Road Anomaly Detection”, “Deep Learning based Road Anomaly Detection”, and “Hybrid Approaches for Road Anomaly Detection”. These approaches are further structured on the basis of model they used, dataset they followed, and accuracy they achieved while detecting road anomalies. Broadly, the researchers used Pothole-600, RAD, and A*3D in their approach, and their performance using parameters such as accuracy, precision, recall, F1-score, IoU, and mAP [5].

III. LITERATURE SURVEY

Road anomaly detection is a crucial area of research and application that focuses on identifying irregularities or damages in road infrastructure, which can pose risks to safety and vehicle performance. Detecting these anomalies, such as potholes, cracks, or uneven surfaces, can help in timely maintenance and enhance the safety and comfort of road users. Below is the classification of approaches for road anomaly detection:

A. Machine Learning based RAD

Road Anomaly Detection using Machine Learning leverages the data-driven techniques to detect anomalies such as cracks, potholes, or other surface irregularities. Leveraging machine learning, Utaminingrum et. al, 2023 employed genetic algorithm for feature selection, and the classification is performed using an Extreme Learning Machine (ELM). Initially, 128 Gray-Level Co-occurrence Matrix (GLCM) features are extracted, then tested in three iterations - 57 features yield an accuracy of 88.65%, whereas combinations of 20 and 12 features scored accuracies of 87.36% and 86.48%, respectively [3]. Later, to facilitate real-time detection, a sliding window technique is implemented by Gao et. al, 2024. They conducted an experiment with four different sites; 3 out of 4 sites range from 71 to 100% accuracy. Otherwise, average road anomaly detection rate is 81.1%, achieved across all the sites. To enhance offline detection capabilities, a high pass filter – optimizer is introduced. This makes the method suitable and adaptable for the real-time operational applications [1]. In contrast to Utaminingrum et. al, 2023, Gao et. al, 2024 focuses on real-time detection of road anomalies.

In comparison to Utaminingrum et. al, 2023 and Gao et. al, 2024, Qunfang et. al, 2024 combines ML with geophysical detection validation in order to reduce the risk of urban road collapses. Initially data augmentation is performed using 3 oversampling techniques including random, SMOTENC, and ADASYN. Then, three ML models SVM, RF, XGBoost are employed in this research to predict road collapse by inducing historical patterns. An impressive

accuracy of 0.9608 and AUC 0.9796 is achieved by combining SMOTENC with XGBoost amongst other algorithms [2]. It confirms the generalizability of XGBoost for real-world anomaly detection [2]. Along with Qunfang et. al, 2024, Zhao et. al, 2024, also focuses on geo-spatial detection and introduces a two-stage ML approach to extract potential anomalies from the dataset and later the anomalies are classified into four categories. Unlike other approaches, it converts time-series data into geospatial to detect road anomalies regardless of vehicle speed. The proposed approach monitors road health by achieving 87% of accuracy [4].

B. Deep Learning based RAD

Unlike ML, Deep learning leverages advanced neural networks instead of data-driven techniques to identify and classify road anomalies such as potholes, cracks, and other surface irregularities by calculating deviations from the normal road conditions. Such techniques are useful to extract features and detect anomalies out of complex data patterns. In response Pustokhina et. al, 2021 introduced deep learning-based anomaly detection technique for pedestrian walkways (DLADT-PW) to ensure the safety of vulnerable roads. The model is primarily aimed identify and classify road anomalies using Mask Region Convolutional Neural Network (Mask-RCNN) with DenseNet and achieved accuracy of 89% across multi-dimensional data [5]. Then, Lee et. al, 2023 used convolutional neural network (CNN) to detect potholes and quantify damage extent using pixel-based approach with an accuracy of 95% [22]. Further, Sabor et. al, 2023 introduced ‘Convolutional-based Mixer Neural Network Road Anomalies Detector and Classifier’ (CMNN-RADC) to detect and classify anomalies. The approach combines low and high-frequency features to accelerate the process. The two-stage process helps to reduce power consumption as well as computational time, and achieved detection rate of 99.3%, and F1 scores up to 99.05% for different road anomalies [6].

Lately, in the approach of Hu et. al, 2024, YOLO-based techniques are chosen to evaluate the real-time performance. This YOLOv5 was designated for vehicle-mounted images and employed Slim-Neck structure to focus on crack through an attention mechanism. Silu activation function and CIoU loss function are utilized to enhance efficiency and accuracy of the model. Finally, an accuracy of 95.6% is achieved with an additional SPPCSPC structure [8]. Then, Paramarthalingam et. al, 2024 uses YOLO for pothole detection and provide real-time auditory or haptic feedback to visually impaired users with an accuracy of 82.7% [25]. In comparison to Hu et. al, 2024, Ren et. al, 2023 introduces a novel approach to detect pavement damage using YOLOv5, incorporates a large-scale detection layer in order to enhance the precision of 78% through a Generalized Feature Pyramid Network (Generalized-FPN) structure. The model is found as significant for road maintenance planning [9]. In comparison to Hu et. al, 2024, Ren et. al, 2023 focused on YOLOv5, a two-stage traffic sign recognition system developed by Hsieh et. al to enhance safety and prevent

serious traffic using YOLOv7. The model achieved an accuracy of 99.2% while training and accuracy of 99% while model testing. The extracted road sign images were rectified with Optical Character Recognition (OCR) tools such as EasyOCR and PaddleOCR. It enhances the capability of self-driving cars to contribute safety by detecting and interpreting traffic signs [10]. In contrast to Hu et. al, 2024 and Ren et. al, 2023, Sabarinathan et. al addresses the demand of efficient road maintenance by proposing YOLOv8-based model. The model can utilize real-time capabilities with and enhanced precision and reliability. The model claims superior accuracy and supports efficient navigation of automated vehicles [19].

C. Hybrid Models for RAD

Hybrid approaches in road anomaly detection combine different techniques or methodologies including machine learning, deep learning, and traditional to leverage their respective strengths [20]. This results in improved performance, robustness, and adaptability to varying conditions and data types. In recent times, the researchers have shifted their focus to mobile devices equipped with motion sensors for low-cost road surface monitoring. Emphasizing the same, Sattar et. Al, 2021 developed a prototype using MATLAB and ArcGIS to conduct a sensor data analysis, and facilitate geocoding and geo-visualization with 87% performance. Meanwhile, threshold-based ML techniques are combined to conduct self-adapting and real-time road anomaly detection and classification [11]. Similarly, Yu et. al, 2023 introduced an ensemble framework using autoencoder to detect road anomalies. It integrated spatiotemporal techniques for graph learning and masking to model local contexts. Meanwhile, random masking was applied to generate heterogenous masked graphs out of input graph. Then, multiple autoencoders are trained on masked graphs to enhance feature learning. It outperforms the tradition ML and DL techniques with an accuracy of 89% [13].

Natha, et. al, 2023 introduced an ensemble model by integrating VGG19, Inception V3, and ResNet50 (three pre-trained CNN models) known as SERAD i.e. Stack Ensemble Road Anomaly Detection. The experimental results proved the model robust and reliable with an impressive accuracy of 98.7%, to detect and classify the road anomalies [21]. In comparison Garay et. al, 2023 proposed an advanced model using transfer learning with Segformer network to detect pothole. The model achieved 86.8% accuracy, 90.01% precision, 90.433% F1 score, and 90.87% Recall [23]. In contrast Xing et. al, 2024 proposed Mask R-CNN model to reconstruct 3D structures of potholes. The model attained accuracy of 82% to calculate damage ratio, with 98% detection accuracy, and 94% calculation accuracy. It ensures surface irregularities' precise identification. Furthermore, for real-time RAD,

Wang et. al, 2024 integrates edge computing with TinyML with enhanced capabilities. For feature extraction, BNN (Binary Neural Network) is used along with 28-fold reduced memory usage and 25% enhanced speed by 2.5% reduction in accuracy [12]. Meanwhile, Han et. al, 2025 uses data fusion and deep anomaly detection for detecting road anomalies with high-emission vehicles with 95% AUC [14].

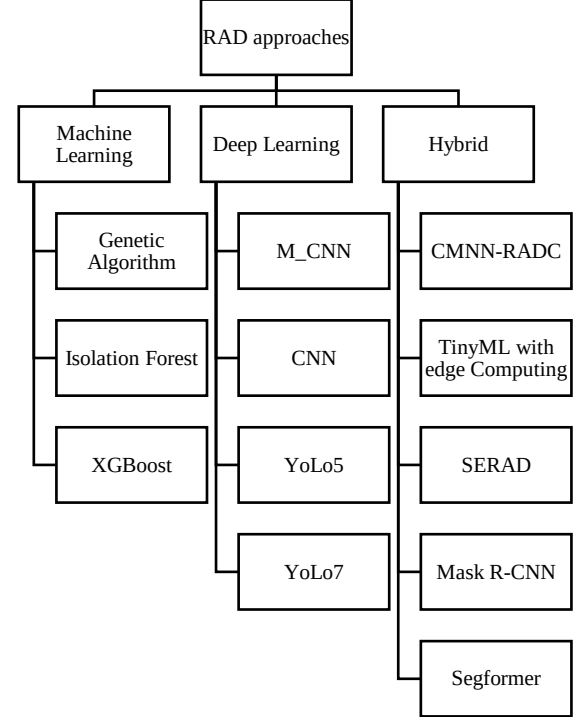


Figure 1: Classification of RAD approaches

IV. LITERATURE GAPS

After conducting comprehensive and comparative literature review, there are certain gaps are identified need to be focused in the upcoming research work. Firstly, it has been discovered that existing approaches are limited to conduct multi-task learning to perform multiple tasks (texture and object anomalies) simultaneously. Such studies would need to focus on unsupervised as well as supervised machine learning. Then, the studies are lagging to focus on dynamic environmental conditions that may affect the accuracy of real-time anomaly detection. Further, the current complex models are working on too many features for which effective feature selection techniques are required. It is important to address potential biases.

Table 1: Comparative Analysis of Approaches for RAD

Author	Algorithm	Approach	Dataset Used	Performance
(Utaminingrum et. Al, 2023)	Genetic Algorithm	Machine Learning	CBIR with 128 Gray-Level Co-occurrence Matrix (GLCM) features	88.65% Accuracy
(Gao et. Al, 2024)	Isolation Forest	Machine Learning	METR-LA	89.6% Accuracy
(Qun Fang et. Al, 2024)	Extreme Gradient Boosting (XGBoost)	Machine Learning	Open Street Map Infrastructure Data	accuracy of 0.9608 and AUC of 0.9796
(Zhao et. Al, 2024)	a two-stage machine learning approach	Machine Learning	Pothole-600 Dataset	87% Accuracy
(Pustokhina et. Al, 2021)	MaskRCNN	Deep Learning	UCSD anomaly detection	89% Accuracy
(Sabor et. Al, 2023)	CMNN-RADC	Deep Learning	Pothole-600	detection rate of 99.3%, and F1 scores up to 99.05
(Ren et. Al, 2023)	YOLOv5 network	Deep Learning	Pothole-600	79.8% Accuracy
(Hu et. Al, 2024)	YOLOv5	Deep Learning	3D-Road	95.6% Accuracy
(Ren et. Al, 2023)	YOLOv7	Deep Learning	Pothole-600	99.2% Accuracy
(Sattar et. Al, 2021)	threshold-based techniques with machine learning approaches	Hybrid	Pothole-600	87% Accuracy
(Yu et. Al, 2023)	ensembled autoencoder	Hybrid	METR-LA	89% Accuracy
(Wang et. Al, 2024)	Integrating Tiny Machine Learning (TinyML) with edge computing	Hybrid	A*3D	90% with 25% increase in inference speed
(Han et. Al, 2025)	data fusion and deep anomaly detection	Hybrid	Pothole-600	95% Accuracy
(Natha, et. al, 2023)	SERAD model	Hybrid	Road Anomaly Dataset - Kaggle	98.7% Accuracy
(Lee et. al, 2023)	CNN model	Deep Learning	UDTIRI	95% Accuracy
(Garay et. al, 2023)	Segformer network	Hybrid	Road Anomaly Dataset – Mendley	86.8% accuracy, 90.01% precision, 90.433% F1 score, and 90.87% Recall
(Xing et. al, 2024)	MASK R-CNN model	Hybrid	Road Anomaly Dataset - Kaggle	detection accuracy of 98% and

				segmentation accuracy of 94%, and calculation accuracy of 82%
(Paramarthalingam et. al, 2024)	YOLO	Deep Learning	Road Anomaly Dataset - Kaggle	image detection accuracy of 82.7%

PROPOSED APPROACH: YOLO-STYLE OBJECT DETECTION MODEL

To conduct new research, a novel approach is proposed for detecting road anomalies. The approach is combining YOLO-style object detection with customized CNN, and claiming to identify texture as well as object-based anomalies in road environments.

YOLO-style object detection model is designed to detect road anomalies in images. It operates by dividing an image into grid of cells. This grid cell is responsible to detect an object, if center of that object falls within it. It further allows the model to make predictions relative to the input image in a single forward pass, and makes it an efficient and suitable for real-time object detection.

This a custom CNN or YOLO-style CNN with convolutional layers that extract spatial features such as edges, textures, and shapes, important for identifying anomalies or objects. Each convolutional layer is followed by ReLU (Rectified Linear Unit), which introduces non-linearity into the model, and sometimes Batch Normalization, to stabilize learning [10].

In short, the model processes road images through convolutional layers, extracts feature and predicts anomalies presence, location, and type. It directly learns to map input images to labeled object detections using a supervised learning approach [13].

A. Proposed Algorithm

Proposed algorithm: RAD using YOLO-style CNN
Input: Road Image F with YOLO-format annotations file L
Output: Object Anomalies Detected
Step 1: Load and preprocess input image as F
Step 2: Parse YOLO annotation file L for each labeled object in L and encode them into grid format
Step 3: Apply transformations to the dataset to make it suitable for YOLO dataset
Step 4: Perform Convolution → BatchNorm → ReLU → MaxPooling
Step 5: Perform forward pass through the model, compute YOLO loss, backpropagate gradients, and update weights using optimizer while model training
Step 6: Perform inference while testing and pass test image through the trained model
Step 7: Display the annotate image with highlighted road anomalies

V. CONCLUSION AND FUTURE DIRECTION

In conclusion, RAD is crucial to maintain road infrastructure quality, enhance traffic safety, and many more. After conducting comprehensive and comparative literature review, ML based approaches are identified as effective to classify the anomalies with higher accuracy. Meanwhile DL can take more complex decisions after extraction and analyzing features using road images. Whereas, hybrid models are combining different models and properties to get more accurate decisions and detect diverse types of anomalies in real-time. In future, the research scholars could focus on advanced machine learning techniques such as ensemble, transfer, and deep reinforcement learning for more accurate and robust results. Moreover, edge computing integrated solutions can be produced for real-time and low-latency road anomaly detection. Furthermore, multi-task learning could be performed to perform multiple tasks simultaneously.

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